

SQL Fundamentals: Select & Filtering

Exercise 01

1. </> SQL

SELECT *

FROM employees;

Employee Table

Id	first-name	last-name	department	Salary	hire-date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2019-03-20	Cape Town
3	Cathy	Zulu	Finance	65000	2018-01-18	Durban
4	David	Mokoena	Marketing	55000	2021-01-05	Pretoria
5	Eva	Mandoo	IT	72000	2017-09-30	Johannesburg

2. </> SQL

SELECT DISTINCT department

FROM employees;

department table

department
IT
HR
Finance
Marketing

3. </> SQL

SELECT first-name, last-name

FROM employees

ORDER BY Salary DESC;



Table

First-name	Last-name
Eva	Naidu
Alice	Green
Cathy	Zulu
David	Indoena
Brian	Lee

4. `</> SQL`

```
SELECT id, first-name, last-name, salary
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

Table

id	First-name	last-name	Salary
5	Eva	Naidu	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

Section 2: WHERE FILTERING

5. `</> SQL`

```
SELECT id, first-name, last-name, department
FROM employees
WHERE department = 'IT';
```

Table

id	First-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidu	IT



6. </> SQL

```
SELECT id, first-name, last-name, department, salary
FROM employees
WHERE department = 'Finance' AND salary > 60000;
```

Table

id	first-name	last-name	department	Salary
3	Cathy	Zulu	Finance	65000

7. </> SQL

```
SELECT id, first-name, last-name, department
FROM employees
WHERE department = 'HR' OR department = 'Marketing';
```

Table

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

8. </> SQL

```
SELECT id, first-name, last-name, department
FROM employees
WHERE department != 'IT';
```

Table

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing



9. 2/7 SQL

SELECT id, first-name, last-name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');

Table

id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Haidan	IT

10. 2/7 SQL

SELECT id, first-name, last-name, department, salary, city
FROM employees
WHERE department = 'IT'
AND salary > 6000
AND city = 'Johannesburg';

Table

id	first-name	last-name	department	salary	city
1	Alice	Green	IT	7000	Johannesburg
5	Eva	Haidan	IT	7200	Johannesburg

